26F3 Homework

Key Ideas and Details and Transformations

What will homework look like?

Every week, we will give you 6–8 questions to work on (about 30 minutes), split between English language arts and mathematics. Some questions will be more challenging than others.

Try your best to submit your work before the start of the next lesson, but remember that homework is optional. You will sometimes see additional practice worksheets. You do not need to complete those; we will only grade the documents labeled "26F*X* Homework."

Homework will be graded out of eight points; a tutor will give you feedback on your work. Feel free to resubmit and correct your mistakes as many times as needed.

English Language Arts: Key Ideas and Details

Adaptation of "Snoozing While Soaring"

1 Researchers have long wondered how some migratory birds can remain in flight for extremely long periods without stopping to rest. In one study, scientists attached small electronic devices to Alpine swifts and discovered that the birds remained airborne for more than 200 days. However, because the devices measured movement rather than brain activity, researchers could not determine whether the birds actually slept while flying.

2 Later, researcher Niels Rattenborg proposed that birds might experience slow-wave sleep while remaining airborne. During this type of sleep, one

hemisphere of the brain can rest while the other remains active. This could allow a bird to continue seeing and navigating while part of its brain sleeps.

3 To test the idea, Rattenborg and his team studied frigatebirds using equipment that recorded brain-wave activity. The researchers found that the birds sometimes entered slow-wave sleep for brief periods while gliding upward on warm air currents. These findings provided stronger evidence that some birds can sleep while remaining in flight.

Adapted from an official NYC Public Schools SHSAT sample passage.

1. Which detail from the passage **best** supports the idea that researchers initially lacked enough evidence to determine whether birds slept during flight? **[1]**
   - **A.** Alpine swifts migrate over long distances.
   - **B.** The electronic devices recorded movement but did not measure brain activity.
   - **C.** Birds can experience more than one type of sleep.
   - **D.** Frigatebirds sometimes glide upward on warm air currents.

   Justify your answer here.

2. How does paragraph 3 contribute to the development of the central idea? **[1]**
   - **A.** It introduces a completely different explanation for bird migration.
   - **B.** It describes an experiment that provides evidence for an earlier hypothesis.
   - **C.** It shows that the first group of researchers incorrectly collected their data.
   - **D.** It explains why frigatebirds migrate farther than Alpine swifts.

   Justify your answer here.

3. Which statement **best** expresses the central idea of the passage? **[1]**
   - **A.** Migratory birds are able to remain awake for hundreds of days without experiencing negative effects.
   - **B.** Scientists gradually gathered stronger evidence showing that some birds can sleep while remaining in flight.
   - **C.** Frigatebirds are better suited for scientific research than other migratory birds.

D. Electronic tracking devices are the most reliable way to study the behavior of birds.

Justify your answer here.

A school introduced an optional morning study program. The principal reported that, as more students began attending the program, the percentage of participating students who arrived to first period on time also increased.

Month	Students Attending Program	Students Arriving to First Period on Time
September	40	75%
October	55	80%
November	70	84%
December	85	88%

4. Which statement **best** explains how the table relates to the principal's statement?

A. The table proves that attending the study program caused every participating student to arrive on time.

B. The table provides numerical evidence showing that greater participation occurred alongside a higher percentage of students arriving on time.

C. The table contradicts the principal because some participating students still arrived late.

D. The table shows that the school required more students to attend the program each month.

Justify your answer here.

Mathematics: Transformations

Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. You may not use a calculator. For all questions in this part, a correct answer with no work shown will receive only **0.5 point**.

5. Compare different statements regarding Figure A and Figure B in the table below. A statement can fulfill one, none, or multiple of the criteria. **[1]**

	Demonstrates $A \cong B$	**Demonstrates** $A \sim B$
Corresponding sides and angles in Figure A **and Figure** B **are congruent.**	No	No
B **is the image of** A **after a translation two units left and a dilation centered at the origin with a scale factor of** $\frac{1}{2}$**.**	No	No
Corresponding angles in Figure A **and Figure** B **are congruent, and corresponding sides in these figures are in proportion.**	No	No
Corresponding angles in Figure A **and Figure** B **are congruent.**	No	No
A **and** B **are polygons that both have the same number of sides and the same number of angles.**	No	No
B **is the image of** A **after a** $180°$ **rotation about the origin and a reflection across the** x **-axis.**	No	No

Justify your answer here.

6. A point (x, y) undergoes a composition of transformations that includes a $-\ 180°$ rotation about the origin, a reflection over the x-axis, and a dilation centered at the origin with a scale factor of $-\frac{7}{2}$. Write the coordinates of the image of the point. **[1]**
   Write your answer here.

△*CDE* contains Points *C*(-1, 6), *D*(7, 5), and *E*(7, 6).
△*FGH* contains Points *F*(5, -1), *G*(10, -1), and *H*(10, -3).
△*JKL* contains Points *J*(1, 0), *K*(-7, -1), and *L*(-7, 0).
△*MNO* contains Points *M*(5, 0), *N*(10, 0), and *O*(10, -2).
△*PQR* contains Point *P*(6, 0), *Q*(-4, 0), and *R*(-4, -4).

7. Identify the two pairs of congruent triangles. For each of these pairs, identify a transformation or composition of transformations that maps one of the congruent triangles onto the other. **[2]**
   Write your answer here.